Forensic Audit: Intuitive Machines IM-2 (Athena), 6 March 2025

A reconstruction from public sources, with theorising on where HATI v2.0 could have intervened

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Foreword: what this document is and isn't

This is desk forensics. I was not inside Intuitive Machines on 6 March 2025. I do not have access to IM's flight recorder telemetry, the post-mission anomaly report, or any internal communications. What I have is what was made public: the SpaceX launch broadcast, IM and NASA press conferences in the days following the landing, tracking data, statements from IM leadership, commentary from planetary scientists active in the space community, and contemporary reporting. Where my reconstruction extrapolates beyond those sources, I say so explicitly.

The IM-2 final mishap investigation report has not been published as of the time of writing (May 2026). When it is, this audit needs a second pass. Until then, what follows is a best-effort reconstruction.

The point of this audit is not to litigate. It is to ask, with the architecture of HATI v2.0 laid out in front of us, a single question: which of Athena's documented or plausible failure modes does v2.0 actually address? Which does it not? I want this question answered before I start writing the Athena counterfactual chapter, not after.

If the honest answer is "two of the five failure modes," that is the result, and the paper has to reflect that. Overreach is not on the table.

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1. Executive summary

On 6 March 2025 at approximately 22:30 UTC, the Intuitive Machines IM-2 lander, call sign Athena, executed a powered descent toward the lunar south pole. The mission target was the Mons Mouton plateau, approximately 84.6°S latitude. Touchdown occurred at a location later confirmed to be on or inside a small crater on the plateau, displaced roughly 250 metres from the intended landing site.

The vehicle came to rest on its side. Power generation through the now-sideways solar panels was insufficient for the planned post-landing operations in a region of permanent low sun elevation. The mission was officially declared complete on 7 March 2025, less than 24 hours after touchdown.

None of the four primary scientific objectives, headlined by NASA's PRIME-1 water ice drilling experiment, could be carried out. Several secondary payloads (the MAPP rover, AstroAnt and Yaoki micro-rovers, the Nokia 4G/LTE demonstration) were unable to deploy.

This is the second consecutive Intuitive Machines lunar landing to end with the vehicle on its side, following IM-1 Odysseus in February 2024. The two failures share architectural similarities and differ in their proximate causes. Athena's failure mode is more clearly tied to perception-layer issues than navigation-layer issues. The perception layer is the half of the problem HATI is built to address. That is why this audit is worth writing.

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2. The vehicle and the target site

2.1 Nova-C class lander

Intuitive Machines' Nova-C is a hexagonal lander, roughly 4 m tall and 2 m across at the base, with six landing legs in a wide stance. Mass on landing is approximately 700 to 900 kg depending on residual propellant. The main propulsion is a single throttleable cryogenic engine using liquid methane and liquid oxygen, supplemented by smaller reaction control system (RCS) thrusters for attitude.

Athena's sensor suite, based on public information:

- Optical navigation cameras for Terrain-Relative Navigation (TRN), matching descent imagery against a pre-loaded LROC-derived basemap.
- A laser altimeter / rangefinder providing altitude above terrain.
- A Doppler lidar (most likely NASA's Navigation Doppler Lidar, NDL, carried as a payload demonstration) providing three-axis velocity.
- An inertial measurement unit (IMU) for attitude and rate sensing.

The lander has a Hazard Detection and Avoidance (HDA) capability in principle, using camera and lidar data during terminal descent to identify hazards and, if necessary, re-target. The specifics of how IM-2's HDA was configured, whether it was active in the final seconds, what thresholds governed re-targeting, are not in the public record I have access to.

2.2 Target site: Mons Mouton

Mons Mouton is a flat-topped massif on the southern lunar limb, between Shackleton crater and de Gerlache crater. It rises approximately 6 km above the surrounding terrain. The plateau surface is among the highest priority targets for water ice prospecting because it borders areas of permanent shadow (where surface ice can persist) while itself receiving sunlight for extended periods (where solar power is viable).

Athena's intended landing coordinates have been publicly stated as approximately 84.6°S, with the longitude in the broader Mons Mouton region (precise number varies between sources). For reference, the Apollo 17 landing site sits at 20°N. We are talking about a fundamentally different lighting and terrain regime than any human mission has flown.

Sun elevation at the landing site during the descent window was estimated below 6° above the local horizon. This is the canonical “low sun angle” regime that makes south polar landing hard. Shadows are long. The dynamic range between sunlit and shadowed terrain saturates standard optical sensors. The contrast between terrain features and their shadows can be misread as terrain features themselves.

This matters for the failure analysis. Every perception system Athena had on board was operating in a lighting regime far outside the conditions under which it was originally characterised. That includes the optical TRN, the HDA, and the visual-spectrum portion of the lidar suite.

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3. Timeline reconstruction

All timestamps in UTC unless noted. Where I am extrapolating between known events, I mark the entry as “est.” or “approx”.

Date / Time (UTC)	**Phase**	**Event**
2025-02-27 00:16	Launch	Falcon 9 lifts off from KSC LC-39A. Payload manifest: IM-2 Athena, plus the separately-targeted Lunar Trailblazer orbiter ride-sharing the same trans-lunar injection burn.
2025-02-27 01:42 (est.)	TLI	Trans-lunar injection complete. Athena separates from the Falcon 9 second stage.
2025-02-28 to 03-03	Cruise	Multiple trajectory correction manoeuvres. IM confirms onboard systems nominal during cruise checks.
2025-03-03 (est.)	LOI	Lunar Orbit Insertion. Athena enters a polar parking orbit.
2025-03-04 to 03-06	Orbital ops	Orbit refinement burns. Sensor calibrations. Final hazard map uplinks.
2025-03-06 21:30 (est.)	DOI	Descent Orbit Insertion. Periapsis lowered to descent altitude.
2025-03-06 22:00 (est.)	PDI	Powered Descent Initiation. Main engine ignites for the braking burn.
2025-03-06 22:15 (est.)	Pitch-up	Pitch-up transition from braking to approach phase. Lateral velocity begins to be killed.
2025-03-06 22:25 (est.)	Terminal	Terminal descent begins. Altitude under 500 m. HDA, if active, operating in this window.
2025-03-06 ~22:30	Touchdown	Touchdown signal received. Vehicle reports contact with the surface.
2025-03-06 ~22:35	First telemetry	Communications established. Initial telemetry suggests an off-nominal attitude.

Date / Time (UTC)	Phase	Event
2025-03-06 ~23:00	Confirmation	IM confirms the lander is on its side. Solar panel orientation incompatible with the post-landing power plan.
2025-03-07 morning	Power analysis	IM announces that battery state and projected solar generation are insufficient for sustained operations.
2025-03-07 afternoon	Mission end	IM officially declares the mission complete. No further attempts to recover scientific objectives.

The window from PDI to touchdown was on the order of 8 to 12 minutes. That is the entire window in which the failure occurred. Most of this audit's forensic interest sits inside that window.

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4. The descent reconstruction

This section is the most reconstructive part of the audit. The second-by-second details of what happened during Athena's descent are not in the public record. What follows is the best reconstruction I can build from IM's public statements in the hours and days after the landing, cross-referenced against what a Nova-C class lander is documented to do during a nominal descent.

4.1 Braking phase (T+0 to ~T+8 minutes from PDI)

The main engine ignites and the lander begins decelerating from orbital velocity. The belly-mounted instruments are tracking the surface for navigation. The TRN system is matching descent camera frames against the pre-loaded basemap. The Doppler lidar provides three-axis velocity measurements. The laser altimeter provides altitude above local terrain.

In principle, the state estimator is fusing all four sources (IMU, TRN, lidar, altimeter) into a single position, velocity, and attitude estimate that the guidance law uses to compute the desired thrust vector.

What IM has confirmed publicly: the descent telemetry during this phase looked nominal until terminal descent. They have not identified any anomalies in the braking phase that contributed to the final failure.

4.2 Approach phase (~T+8 to ~T+10 minutes from PDI)

Pitch-up manoeuvre. The lander rotates from a high-pitch-angle thrust profile to near-vertical. Horizontal velocity is killed off. Vertical velocity is brought into the terminal range. Altitude at the end of this phase is typically a few hundred metres.

What IM has confirmed: nothing specific about this phase. Public statements jump from "descent was nominal" to "touchdown was off-nominal" without much in between.

4.3 Terminal phase (~T+10 to T+12 minutes from PDI)

Under 500 m altitude. The lander is now vertically oriented, descending at a few metres per second. The TRN should now be providing very tight position updates, because the descent camera is looking at a small, high-resolution patch of terrain very close to the touchdown point. The HDA, if active, should be scanning this immediate touchdown footprint for hazards (slopes, rocks, craters) and recommending re-target if necessary.

This is the phase where the failure happened. The lander touched down approximately 250 m laterally displaced from its intended target. It touched down on a slope inside a small crater. It touched down with enough lateral velocity that one or more landing legs collapsed.

Each of these three observations corresponds to a different candidate failure mode. Section 6 takes them one by one.

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5. The terrain at touchdown

The actual landing site has been observed post-event by LRO. Public imagery from the LROC team at Arizona State shows Athena lying on its side on or just inside the rim of a small crater. The crater's estimated diameter is 10 to 20 metres, with a depth of a few metres.

The crater is not visible as a discrete feature in the pre-mission basemap at the resolution Athena's TRN was using for matching.

That last sentence is the one that matters for HATI v2.0.

A crater this size sits squarely in the resolution gap. At 1.5 m/px DEM resolution it is right at the Nyquist boundary, where the rim might imprint as roughness but not as a detectable feature. At the basemap resolution Athena was matching against, the crater is below detection. It exists as roughness in the topographic statistics, not as a discrete labelled hazard.

We do not have public access to IM's pre-mission hazard map for Mons Mouton. We can only theorise on whether this specific crater appeared as a hazard on their map. Based on the public reporting of the landing's displacement and final attitude, it appears it did not, or the threshold IM had set for active avoidance was above this crater's severity classification.

This is, structurally, exactly the failure mode HATI v2.0's heatmap is designed to address.

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6. Candidate failure modes

These are not mutually exclusive. The actual failure was almost certainly a combination of several. Listing them separately is for the purpose of analysis and intervention mapping.

F1. Sub-resolution hazard not flagged

The landing site contained a hazard (the small crater) that was not represented in the pre-mission hazard map at the resolution the lander was using for autonomous decision-making. The HDA either did not detect it during terminal descent or detected it too late to act.

Evidence: post-event LROC imagery confirms the crater. The crater is small. Sub-resolution hazards are exactly what HATI's heatmap is built to flag.

F2. Optical TRN compromised by lighting conditions

The optical TRN system matches descent camera frames against a pre-loaded basemap. Under low sun angles, the descent imagery has dramatically different shadow patterns than the basemap (which was acquired at different sun angles). This degrades the matching confidence. If TRN confidence dropped below the threshold where the state estimator weighted it heavily, the lander may have been navigating partly on IMU-only data with no position correction in the final seconds.

Evidence: IM-1 had documented optical TRN issues at much higher latitudes. South polar low-sun-angle is a substantially harder regime. IM has publicly acknowledged that the lighting environment was a significant challenge. The 250-metre landing displacement is consistent with a degraded position estimate in the final seconds.

F3. Altimeter or lidar performance degraded

Athena's altimeter and Doppler lidar may have produced noisier or biased measurements over the highly reflective sunlit terrain interspersed with deep shadow (effective "black hole" pixels from a lidar's perspective). A biased altitude estimate would cause the lander to under- or over-shoot the desired touchdown altitude.

Evidence: IM-1 had documented laser altimeter problems (the safety pin issue). IM-2's altimeter was, by all accounts, functional, but operating in a much harder regime than IM-1. The 30-degree-plus tilt at touchdown could be partly attributable to an altitude bias that triggered touchdown shutdown of the main engine while the lander was still moving laterally.

F4. Insufficient propulsion margin or attitude control authority

If the lander's RCS thrusters did not have enough authority to kill lateral velocity at the rate the guidance law demanded, or if the main engine throttling profile was not properly bracketing the descent rate, the lander may have arrived at the surface with more lateral velocity than its landing gear could absorb.

Evidence: the tipped-over outcome is consistent with too much lateral velocity at touchdown. IM has publicly mentioned propulsion-related discussion in their post-event communications. Details are not in public view.

F5. HDA threshold or re-targeting authority too conservative

Even if the HDA did detect the crater rim or the slope late in descent, it may have lacked the propellant margin or the timeline to execute a useful re-target. Or its thresholds may have been set too conservatively to trigger at all.

Evidence: 250 m displacement could be consistent with a late re-target attempt that didn't have enough authority to clear the original hazard zone but did move the lander into an alternate (also hazardous) zone.

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7. Where HATI v2.0 could have intervened

For each of the five candidate failure modes, here is the layer of v2.0 that addresses it, and an honest assessment of how strong the intervention would be.

F1 (sub-resolution hazard not flagged) → HATI heatmap

This is the strong case. The heatmap is built specifically for sub-resolution hazards.

A patch of terrain containing a 10 to 20 metre crater is exactly the kind of patch the heatmap channels are designed to detect through statistical roughness, even when the crater itself doesn't appear as a discrete feature in the DEM. The MDS at L = 10 pixels would see the curvature signature from the rim. The Hurst exponent would see the excess fine-scale structure. The IQR of profile curvature would see the rim-to-floor curvature contrast even when the rim height is below detection.

If the heatmap had been generated for the Mons Mouton plateau before Athena's launch, and if IM's hazard catalogue had been augmented with it, this specific crater's location would almost certainly have been flagged as a high-heatmap region.

How strong is this intervention? Conditional. The heatmap would have provided the information. Whether IM's HDA system would have weighted that information appropriately, whether the re-targeting authority would have respected it, whether the descent corridor would have been adjusted in advance based on the heatmap, are all separate questions about IM's system, not HATI's. The heatmap is a perception layer. It feeds decisions; it doesn't make them.

This is still the case where HATI v2.0 most plausibly intervenes. It is the load-bearing claim of the Athena counterfactual.

F2 (optical TRN compromised by lighting) → HATI v2.0 TRN

This is where the v2.0 TRN proposal earns its keep.

IM's optical TRN works by matching whole image frames against a basemap. That approach is fragile under lighting changes. The basemap was acquired at one set of sun angles. The descent imagery is at another. Shadows fall differently. Albedo contrasts shift. Matching confidence degrades.

HATI v2.0 TRN works differently. Scout's YOLO detects discrete features (boulders, craters) in the descent imagery. Titan's offline pass detects the same kinds of features in the basemap, producing a landmark database. The matching is between detection sets, not between raw image patches. Shadow patterns are not the matched quantity. Feature positions are.

Boulders and craters cast shadows that move with sun angle, but the boulder and the crater are the same boulder and the same crater regardless of when you image them. A detection-based TRN should be more robust to lighting changes than image-template TRN by construction.

How strong is this intervention? Architecturally plausible. The TRN has not yet been implemented or validated, so I can't quote numbers. The v2.0 development plan includes a TRN validation section specifically targeting low-sun-angle robustness because of Athena. If we hit the target convergence rate (>= 90% of synthetic descent frames) in the realistic lighting regime, the v2.0 TRN would have given Athena a more reliable position estimate during the seconds in which IM's TRN was struggling.

F3 (altimeter/lidar performance) → not addressed by HATI

HATI's perception layers do not measure altitude. The Scout YOLO detects features in images; it does not produce a metric altitude estimate. The heatmap is a static map; it does not provide real-time state estimation.

If Athena's altimeter was producing biased readings, no version of HATI fixes that.

This is the failure mode where I have to be most honest about HATI's scope. We are a perception system. We are not a sensor suite redesign.

F4 (propulsion margin / attitude control) → not addressed by HATI

Same. If the main engine throttling profile or the RCS authority was inadequate, HATI does not intervene. The LM-7 6-DoF simulator in HATI v1.5 models a specific (Apollo-derived) propulsion system. An Athena counterfactual run would have to substitute Nova-C's actual propulsion model to be meaningful. That is doable, but it is propulsion modelling, not perception modelling.

F5 (HDA threshold / re-targeting authority) → addressed structurally, not at the proximate trigger

This is the case where HATI v2.0's broader architectural argument bites.

If Athena's HDA was reactive (scanning during terminal descent, deciding in real time whether to re-target), and if the decision arrived too late to act on, then the proximate problem is the timeline of HDA itself. HATI's architectural answer is that the heavy hazard analysis should be done in advance, on the ground (Titan), so the in-flight HDA can be a confirmation step rather than a discovery step.

If the Athena crater had been on a v2.0 heatmap before launch, the descent corridor would not have been targeted there in the first place. The HDA would not have needed to discover the hazard mid-flight.

This is the strategic intervention. It is also the most diffuse, because it presumes ground-truth changes to the mission planning pipeline, not just the in-flight stack.

Summary

Failure mode	v2.0 layer that addresses it	Strength
F1 Sub-resolution hazard not flagged	Heatmap	Strong (conditional on integration)
F2 Optical TRN lighting failure	v2.0 TRN	Architecturally plausible, not yet validated
F3 Altimeter/lidar bias	Not addressed	None
F4 Propulsion / attitude control	Not addressed	None
F5 HDA timeline	Mission planning (heatmap as input)	Strategic, indirect

Out of five candidate failure modes, HATI v2.0 directly addresses two (F1, F2), partially addresses one (F5), and does not address two (F3, F4).

This is the honest assessment. The two we address are the perception-layer failures. The two we do not are sensor and propulsion failures. That is the scope of the v2.0 paper, and it should be the scope of the Athena counterfactual chapter.

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8. What this means for the counterfactual run

When I'm back at the GPU box, this audit's outputs feed the counterfactual experimental design as follows.

Inputs to the counterfactual

The LROC DEM tile centred on the actual touchdown coordinates with a window large enough to cover plausible divert footprints. 5 km by 5 km based on the displacement statistics in HATI v1.5.

The LROC NAC multi-illumination stack for the same region. Mons Mouton has dense NAC coverage at varying sun angles, which is the input the v2.0 shadow channels need.

The IM-2 published touchdown point and the announced intended landing point. We want the offset between them as a real-world calibration of how far off-target the existing TRN was.

What the counterfactual measures

A Monte Carlo of 100 (or 500, for tighter confidence intervals) simulated landings on the Mons Mouton terrain, with HATI v2.0 active versus disabled. The success metric: did the lander avoid the specific small crater on which Athena tipped, and did it touch down within the gear's lateral velocity tolerance?

What the counterfactual cannot measure

Whether HATI v2.0 would have prevented Athena's actual failure. We do not have IM's flight code. We do not have their actual sensor traces. We can only ask: would a v2.0-equipped lander, on the same terrain, with realistic noise on the same class of sensors, have hit the same crater? If the heatmap had flagged it, the divert logic should have routed around it.

The honest framing of the result: "On the Mons Mouton terrain under realistic noise, a HATI v2.0-equipped lander avoided the touchdown-region hazards that brought down IM-2 in X% of 100 simulated descents, 95% CI A-B%."

That is the publishable claim. Anything stronger overreaches.

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9. What this audit changes about v2.0 development priorities

Writing this changed a few things about how I think about v2.0 development:

The low-sun-angle robustness of the v2.0 TRN is no longer a nice-to-have. It is the specific differentiator that makes the Athena counterfactual relevant. The TRN validation set has to include synthetic descent frames at sun elevations matching the actual descent conditions (below 6°). If the TRN works at 30° elevation and falls apart at 6°, the Athena counterfactual claim is empty.

The NAC shadow channels in the heatmap also need to be validated specifically under polar low-sun conditions. A 50 cm boulder casts a 5 m shadow at 6° sun elevation, yes. But what about a 10 m crater? Its shadow geometry is more complex (the crater rim shadows its own floor, which is itself shadowed by something on the other rim). The validation set for NAC shadow stats should include south polar imagery, not just equatorial or mid-latitude.

The forensic audit data set (the 1,459-object v1.5 audit) was collected on the Apollo 17 site, at 20°N. Mid-latitude. Sun elevation during the imagery was much higher than Mons Mouton. We need to confirm that the heatmap weights, anchored to literature on lunar roughness that itself is statistically dominated by mid-latitude data, transfer cleanly to the polar lighting regime. If they do not, we have a domain shift problem that needs to be addressed before the Athena counterfactual.

The HDA threshold question is interesting but outside scope. I want to mention it in the discussion section of the paper, not investigate it as part of v2.0 work.

The two failure modes we do not address (F3 altimeter, F4 propulsion) need explicit acknowledgement in the Athena counterfactual chapter. The reader will ask. The honest answer: those are real, those are not what we address, those need a different system.

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10. Outstanding uncertainty

What I do not know that I want to know:

The actual sun azimuth and elevation at Athena's touchdown moment, with precision. I have approximate values. The simulator needs better.

IM's actual descent trajectory in the final 30 seconds. Public reporting gives the start and the end, not the middle.

Whether Athena's HDA was actually active during the terminal phase, and what its threshold settings were. Presumably in IM's internal records. Not public.

The exact dimensions of the crater where Athena landed. LROC has imaged it but I do not have a published photogrammetric measurement.

The sensor calibration history for Athena's optical TRN under polar conditions. IM did pre-mission characterisation; the details are not public.

The IM-2 final mishap investigation report itself.

When any of these become publicly available, this audit needs an update.

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11. Sources

This reconstruction draws on:

- SpaceX launch broadcast, 2025-02-27.
- Intuitive Machines press conferences, 2025-03-06 to 2025-03-08.
- NASA Artemis Daily updates, 2025-03-06 onwards.
- Lunar Reconnaissance Orbiter Camera (LROC) post-landing imagery published by the LROC team at Arizona State University.
- Contemporary reporting by Spaceflight Now, Ars Technica, SpaceNews, and the BBC during the week following the landing.
- Independent technical commentary published on space community platforms during March 2025.

No proprietary IM data was used. All conclusions are reconstruction from publicly observable information. Where I have inferred or extrapolated, I have said so.

Closing note

This is a working document. It will be revised when IM publishes the final mishap report, when better LROC photogrammetry becomes available for the landing site, and when the v2.0 development reaches the point where the counterfactual run can actually be performed.

For now, this is the best forensic reconstruction I can build from where I sit. The conclusion I take into v2.0 development: two of the five candidate failure modes are addressable by what I am building, one is partially addressable, and two are not. The Athena counterfactual chapter has to be scoped to that reality.

If anyone reading this has corrections, pointers to public sources I have missed, or technical disagreements with the failure mode analysis, I want to hear them.

— Gergő Csaba Morvai 2026-05-19